Vacum Cleaner Agent

```python
# Input
location = input("Enter initial location (A/B): ").upper()
A = input("Enter Room A status (Clean/Dirty): ").capitalize()
B = input("Enter Room B status (Clean/Dirty): ").capitalize()

print("\nInitial State:")
print("Location:", location)
print("Room A:", A)
print("Room B:", B)

# Vacuum Cleaner Simulation
while True:

    if location == 'A' and A == "Dirty":
        action = "SUCK"
        A = "Clean"

    elif location == 'B' and B == "Dirty":
        action = "SUCK"
        B = "Clean"

    elif A == "Dirty":
        action = "LEFT"
        location = 'A'

    elif B == "Dirty":
        action = "RIGHT"
        location = 'B'

    else:
```

```
    else:
        action = "NO-OP"

    print("\nAction:", action)
    print("Location:", location)
    print("Room A:", A)
    print("Room B:", B)

    if action == "NO-OP":
        break

# Final State
print("\nFinal State:")
print("Location:", location)
print("Room A:", A)
print("Room B:", B)
print("Both rooms are Clean")
```

Output:

```
Enter initial location (A/B): A
Enter Room A status (Clean/Dirty): clean
Enter Room B status (Clean/Dirty): dirty

Initial State:
Location: A
Room A: Clean
Room B: Dirty

Action: RIGHT
Location: B
Room A: Clean
Room B: Dirty

Action: SUCK
Location: B
Room A: Clean
Room B: Clean

Action: NO-OP
Location: B
Room A: Clean
Room B: Clean

Final State:
Location: B
Room A: Clean
Room B: Clean
Both rooms are Clean
```

```
PS C:\Users\n6787\OneDrive\Desktop\c> python -u "c:\Users\n6787\OneDrive\Desktop\c"
Enter initial location (A/B): A
Enter Room A status (Clean/Dirty): dirty
Enter Room B status (Clean/Dirty): dirty

Initial State:
Location: A
Room A: Dirty
Room B: Dirty

Action: SUCK
Location: A
Room A: Clean
Room B: Dirty

Action: RIGHT
Location: B
Room A: Clean
Room B: Dirty

Action: SUCK
Location: B
Room A: Clean
Room B: Clean

Action: NO-OP
Location: B
Room A: Clean
Room B: Clean

Final State:
Location: B
Room A: Clean
Room B: Clean
Both rooms are Clean
```

```
Enter initial location (A/B): B
Enter Room A status (Clean/Dirty): dirty
Enter Room B status (Clean/Dirty): dirty

Initial State:
Location: B
Room A: Dirty
Room B: Dirty

Action: SUCK
Location: B
Room A: Dirty
Room B: Clean

Action: LEFT
Location: A
Room A: Dirty
Room B: Clean

Action: SUCK
Location: A
Room A: Clean
Room B: Clean

Action: NO-OP
Location: A
Room A: Clean
Room B: Clean

Final State:
Location: A
Room A: Clean
Room B: Clean
Both rooms are Clean
```

```
Enter initial location (A/B): B
Enter Room A status (Clean/Dirty): dirty
Enter Room B status (Clean/Dirty): clean

Initial State:
Location: B
Room A: Dirty
Room B: Clean

Action: LEFT
Location: A
Room A: Dirty
Room B: Clean

Action: SUCK
Location: A
Room A: Clean
Room B: Clean

Action: NO-OP
Location: A
Room A: Clean
Room B: Clean

Final State:
Location: A
Room A: Clean
Room B: Clean
Both rooms are Clean
```